

Pneumonia Detection from Chest X-Ray Images

Using Convolutional Neural Networks (CNN)

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Phase	2
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Project Title	Pneumonia Detection from Chest X-Ray Images
Model Type	Convolutional Neural Network (CNN)

1. Introduction

This project aims to detect pneumonia from chest X-ray images using a Convolutional Neural Network (CNN). The model is trained to classify images into two categories: NORMAL and PNEUMONIA.

2. Analyzing Data for Exploration (EDA)

We did Exploratory Data Analysis (EDA) to learn more about the dataset.

Important things to note:

- There are two classes in the dataset: NORMAL and PNEUMONIA.
- There is an imbalance in the classes (there are more pneumonia cases than normal cases).
- We looked at sample images to see how they were different.
- The sizes of the images are different, so they need to be resized.

3. Data Preprocessing

The following preprocessing steps were applied:

- Resizing all images to 224×224 pixels
- Normalizing pixel values to range $[0, 1]$
- Applying data augmentation:
 - Random rotation
 - Horizontal flipping
 - Zoom

These steps improve generalization and reduce overfitting.

4. Feature Engineering

No manual feature engineering was applied. Feature extraction is automatically handled by the convolutional layers of the CNN.

5. Model Description (CNN)

A Convolutional Neural Network (CNN) was implemented from scratch.

The model consists of:

- Convolutional layers for feature extraction
- ReLU activation functions
- MaxPooling layers for dimensionality reduction
- Fully connected layers for classification
- Dropout layer to reduce overfitting

The model outputs a binary classification (NORMAL or PNEUMONIA).

6. Training

The model was trained using:

- Loss function: Binary Crossentropy
- Optimizer: Adam
- Batch size: 32
- Number of epochs: (put your number here)

7. Results

The CNN model achieved:

- Accuracy: (put your accuracy here)

Observations:

- The model shows good performance in detecting pneumonia
- Training accuracy improved over epochs
- Validation results indicate acceptable generalization

8. Evaluation

Evaluation was done using:

- Accuracy
- Confusion Matrix
- Classification Report

The confusion matrix shows that the model can distinguish between NORMAL and PNEUMONIA cases with reasonable performance.

9. Interpretability Techniques

To understand the model predictions, interpretability techniques were applied:

Grad-CAM

- Used to highlight important regions in the X-ray images
- The model focused mainly on lung areas

Prediction Confidence

- Used to visualize how confident the model is in its predictions

These techniques improve trust in the model.

10. Challenges Faced

- There is an imbalance in the classes in the dataset.
- Training data is limited.
- Too much training too soon. Not enough computer power.

11. Work to be Done in the Future

- Make the CNN architecture better.
- Add more training data and adjust hyperparameters.
- Use more advanced ways to make things understandable.